# Classification: Performance Metrics

Chapter 3

# Classification: Accuracy

- The task of labeling data instances as belonging to a particular class
- How good is a ML classification model that we trained
- Accuracy:
  - Number of corrected predictions / total number of predictions

- Is accuracy a good performance metric of a classifier?

# Classification: Accuracy

- Example:
  - MNIST image database of handwritten digits from 0 to 9

70 000 samples total

# Classification: Accuracy

- Example continued:
  - Let us build a binary classifier with two classes:
    - Images representing digit 5
    - Everything else
  - An SGDClassifier with 3-fold cross-validation:

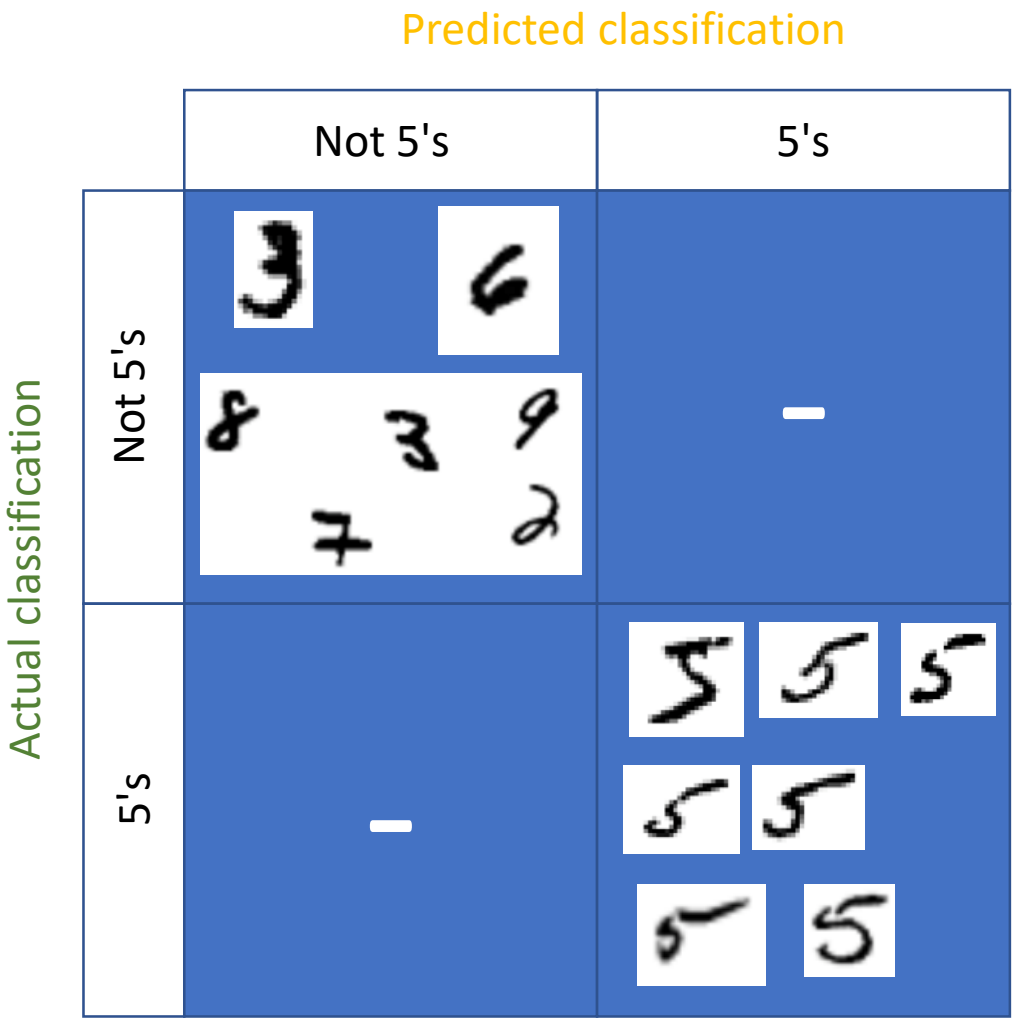
    - Accuracy = 93%

    - Is it good?

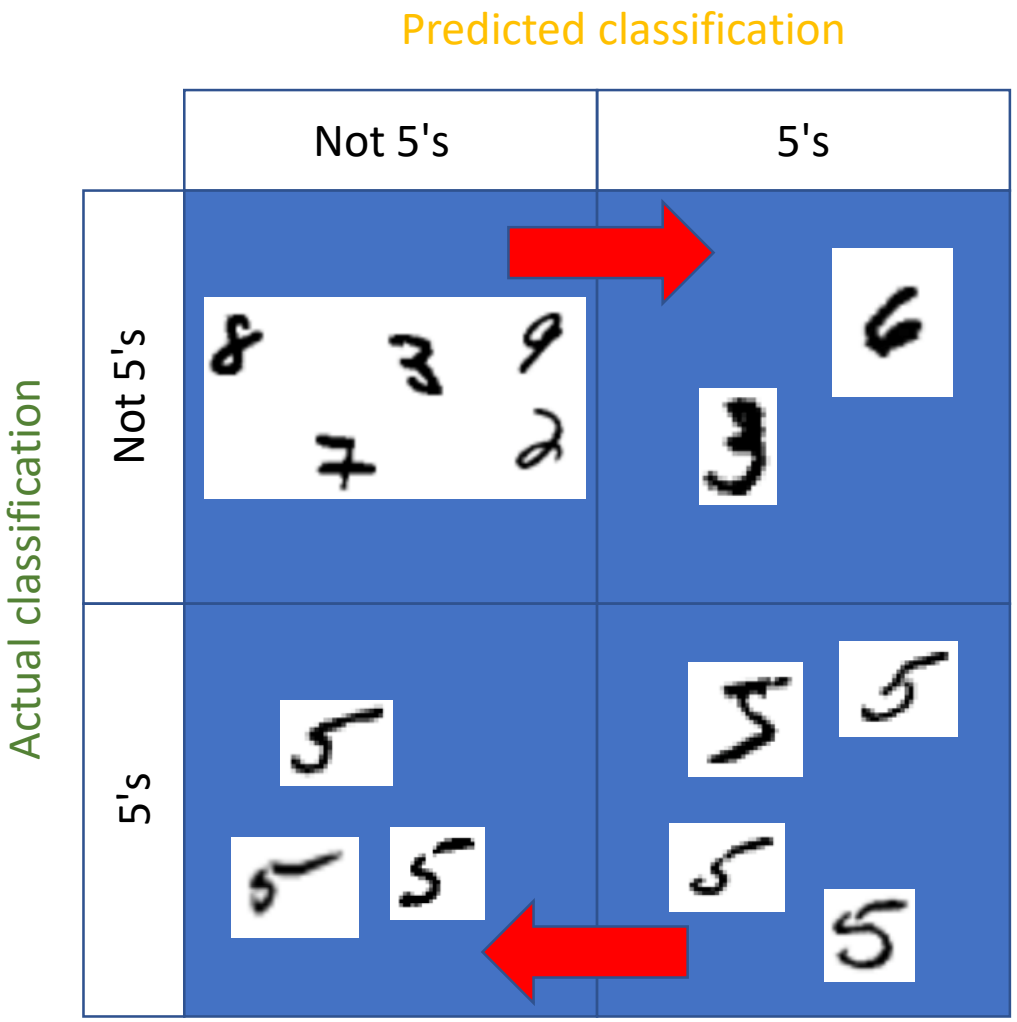
# Classification: Accuracy

- Example continued:
  - Naïve classifier:
    - Label all instances as not-fives
    - Accuracy: 90% !

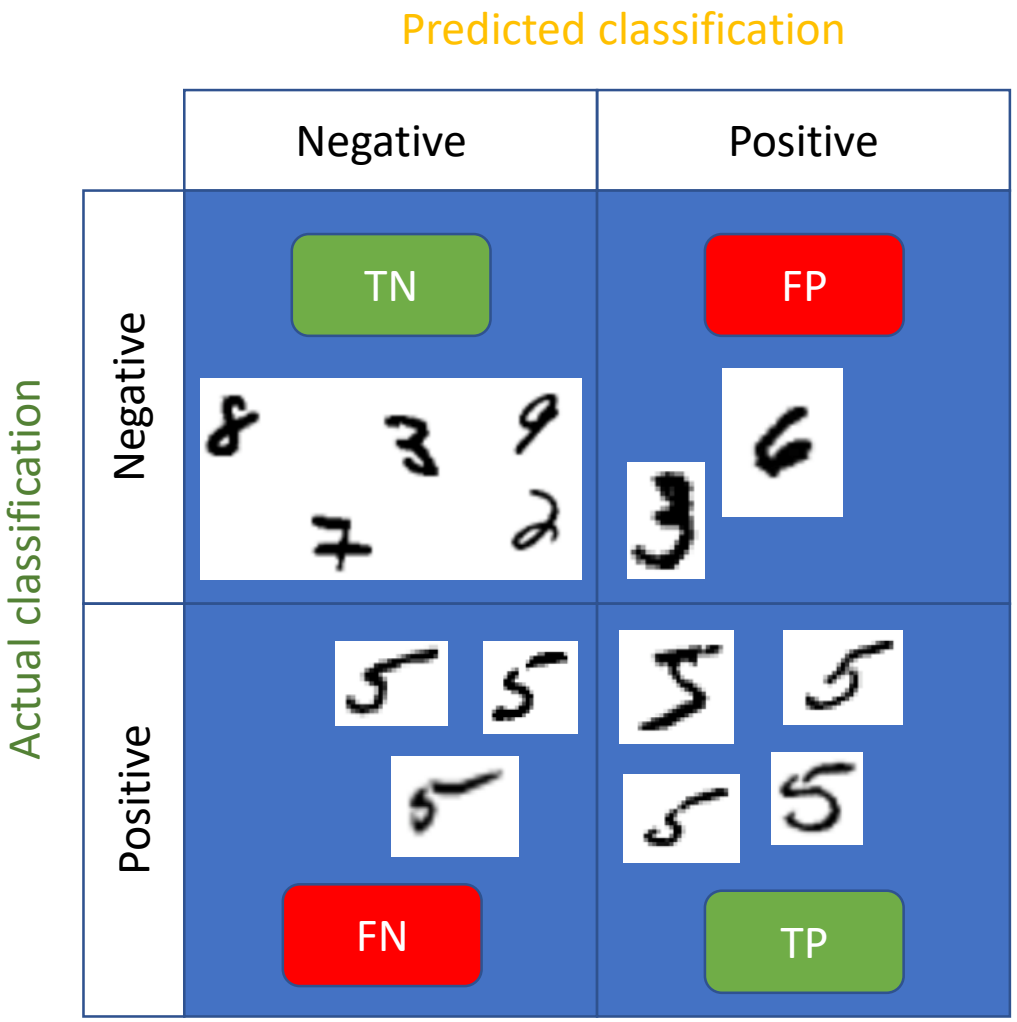
# Confusion Matrix: Perfect Classifier



# Confusion Matrix: Realistic Classifier

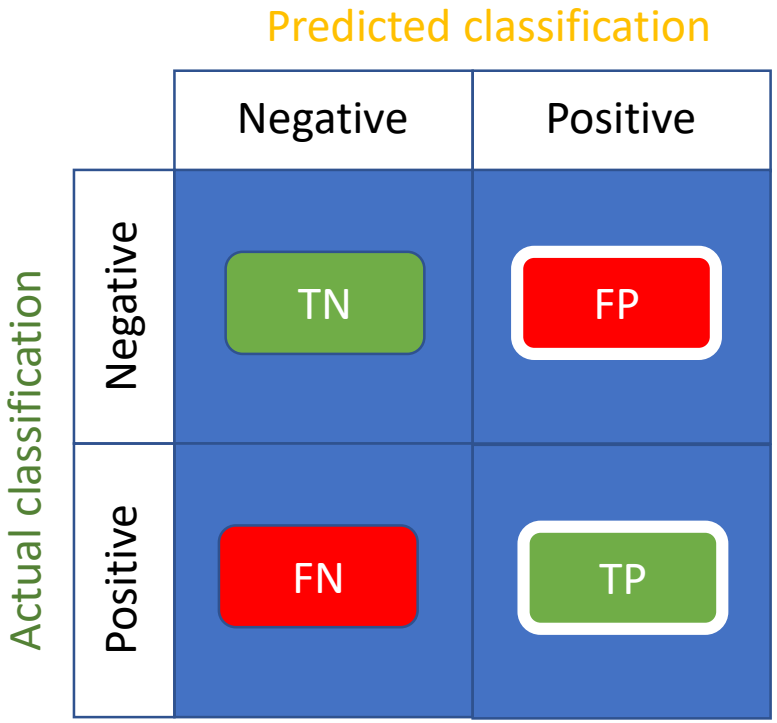


# Confusion Matrix: Terminology



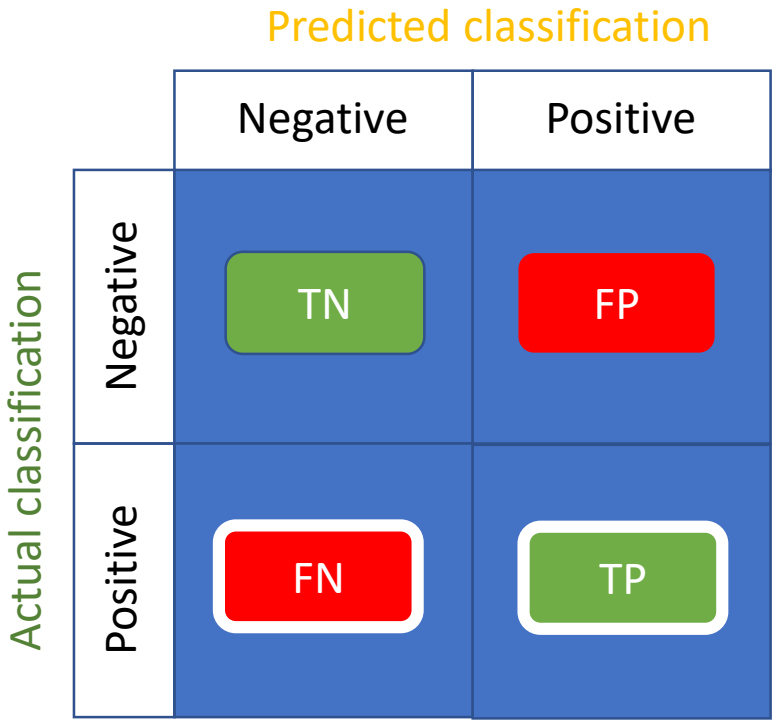
# Confusion Matrix: Precision

$$\text{precision} = \frac{TP}{TP + FP}$$



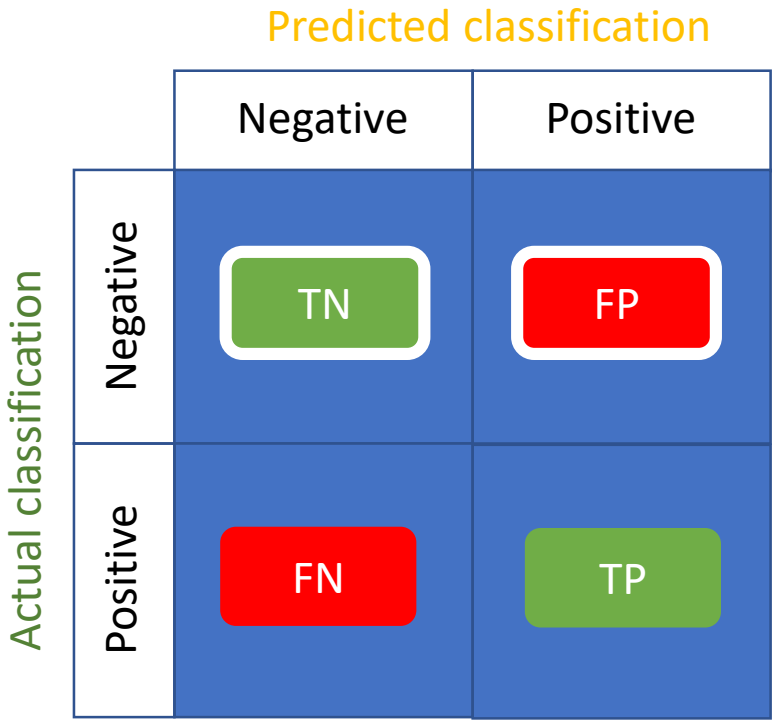
# Confusion Matrix: Recall (a.k.a. sensitivity, True Positive Rate (TPR))

$$\text{recall} = \frac{TP}{TP + FN}$$



# Confusion Matrix: FPR (False Positive Rate)

$$FPR = \frac{\text{FP}}{\text{FP} + \text{TN}}$$



# Precision and Recall

- SGDClassifier
  - Accuarcy = 93%

  - Precision = 73%
  - Recall = 76%

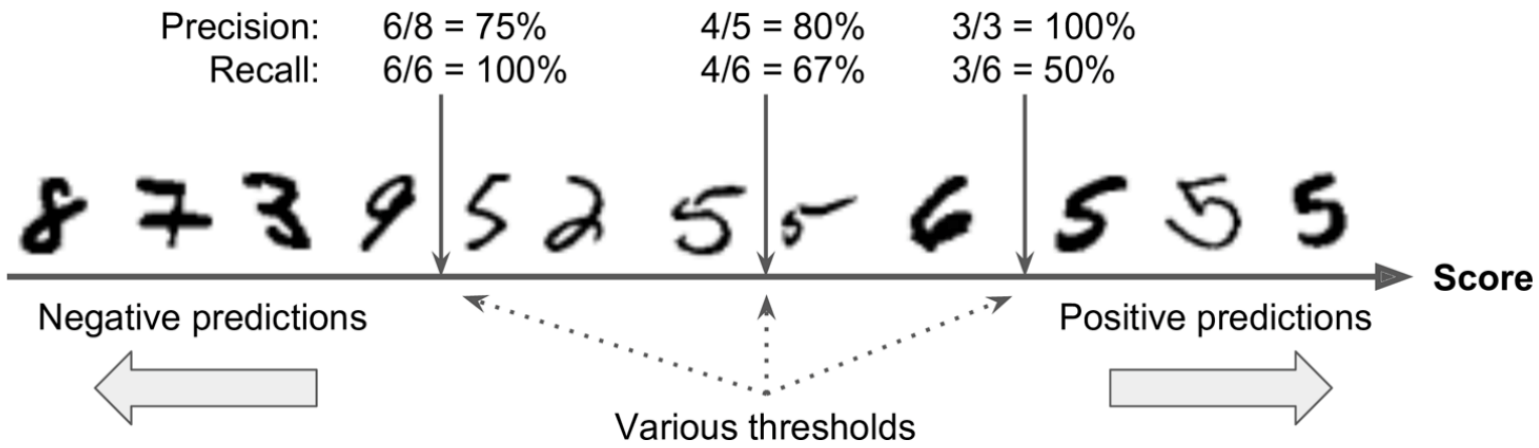
# Precision and Recall: F1 Score

$$F_1 = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{TP}{TP + \frac{FN+FP}{2}}$$

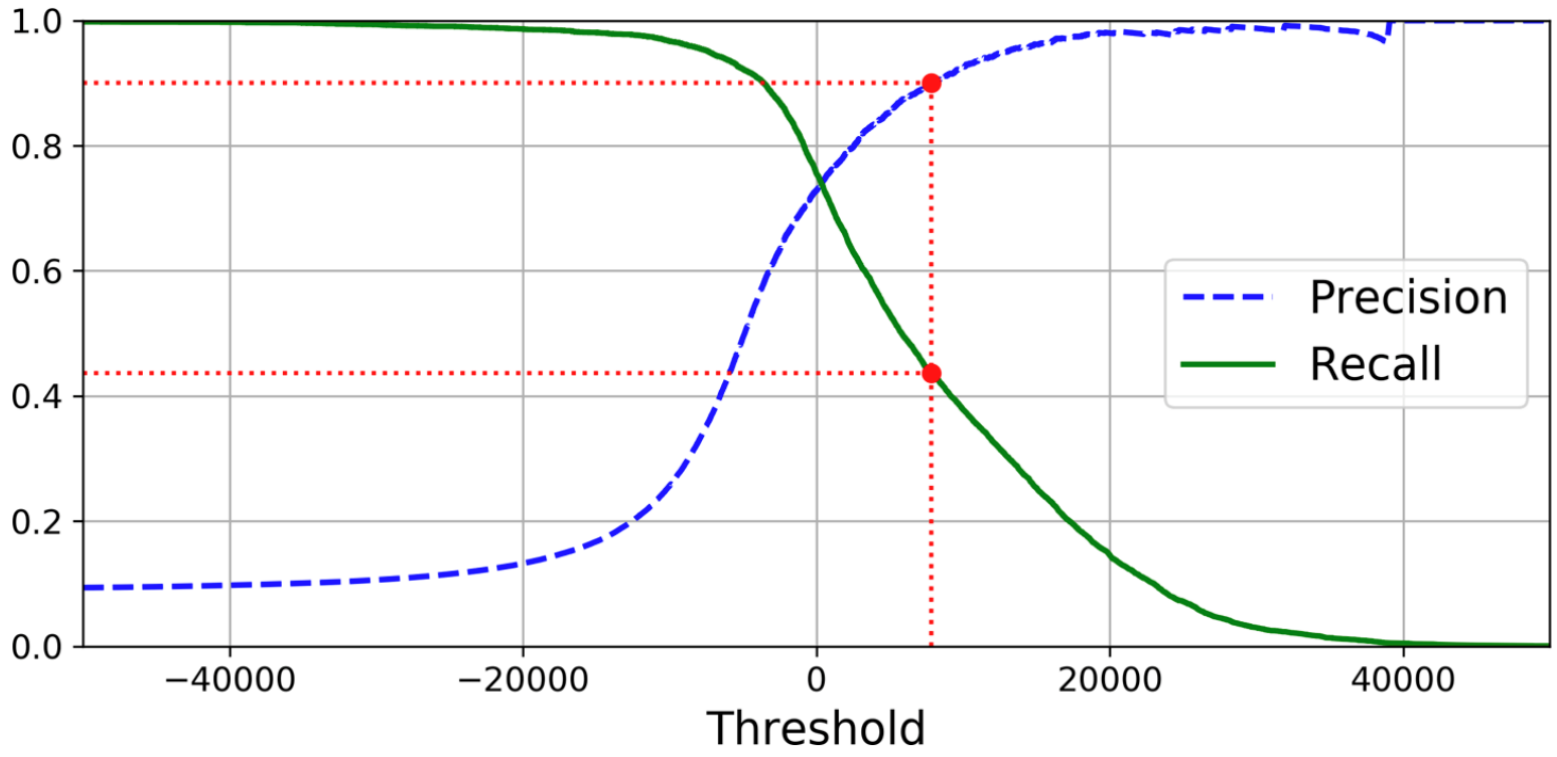
F1 Score favors classifiers with high both precision and recall

But it is not always what is needed

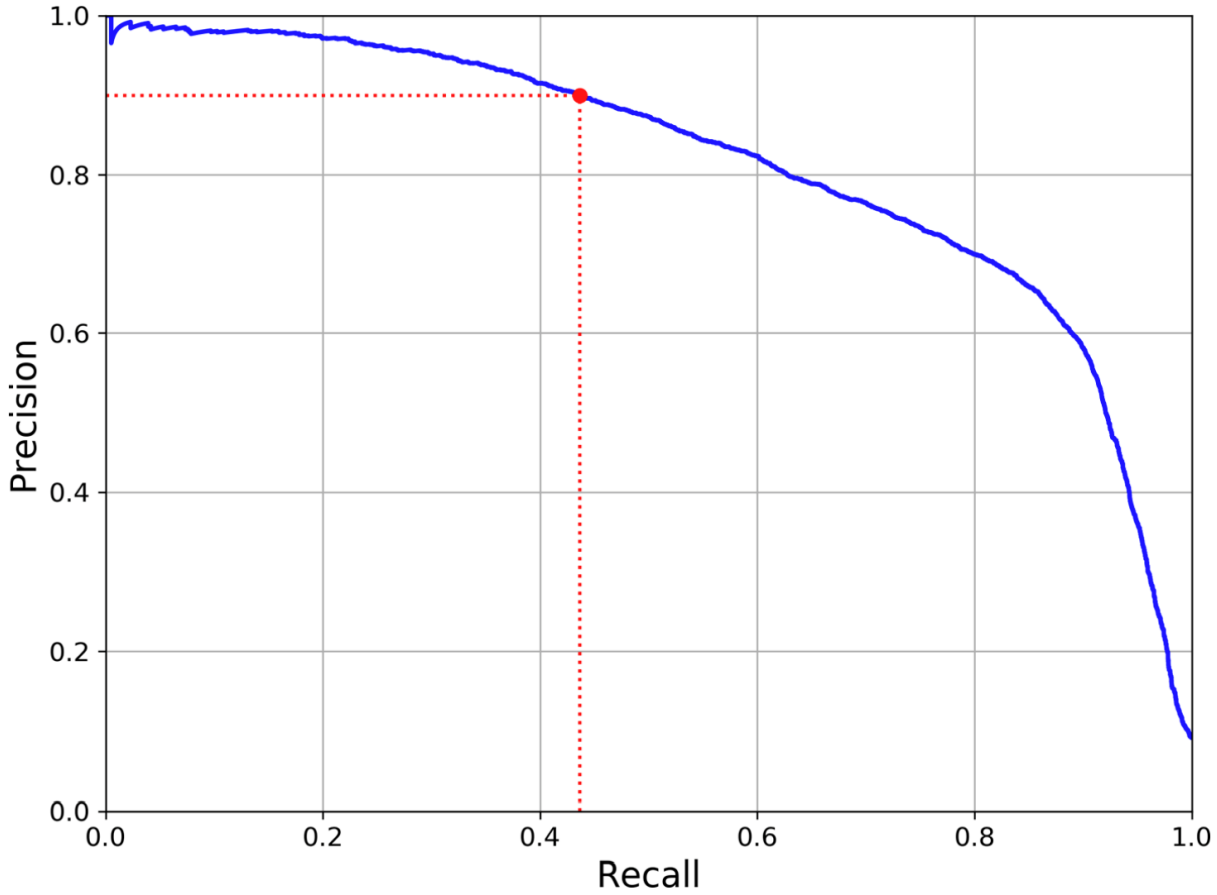
# Precision, Recall and The Decision Threshold



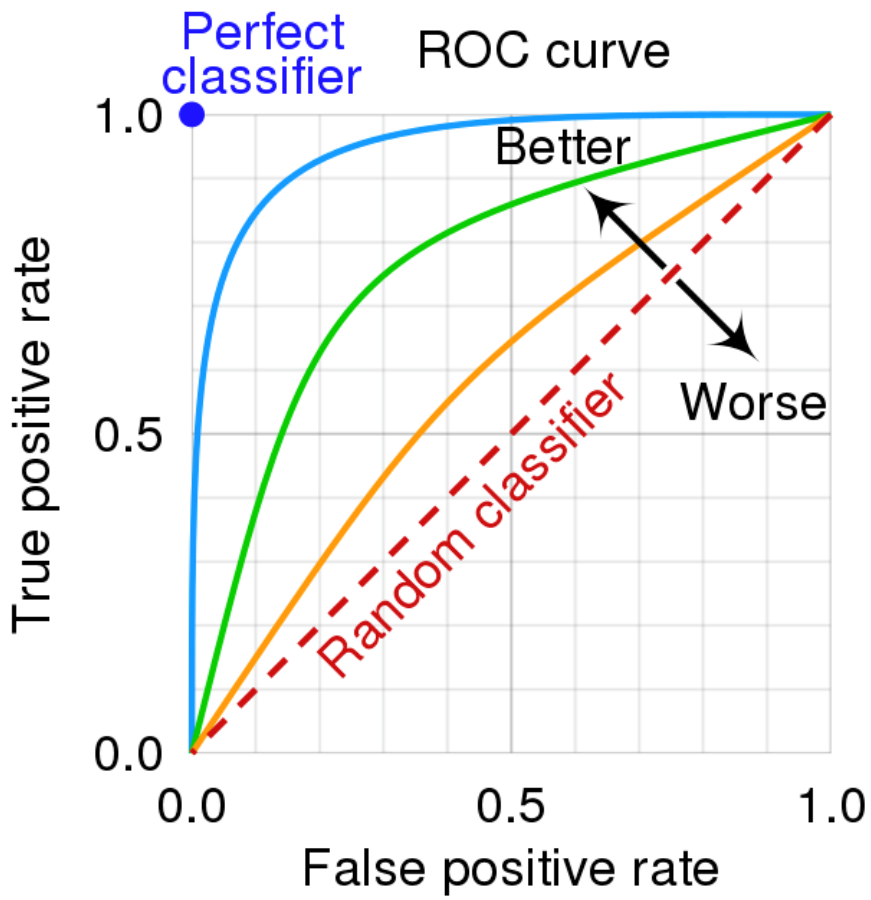
# Precision, Recall and The Decision Threshold



# Precision vs Recall



# ROC: Receiver Operating Characteristic




Source: https://www.wikiwand.com/en/Receiver_operating_characteristic

$$FPR = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$$

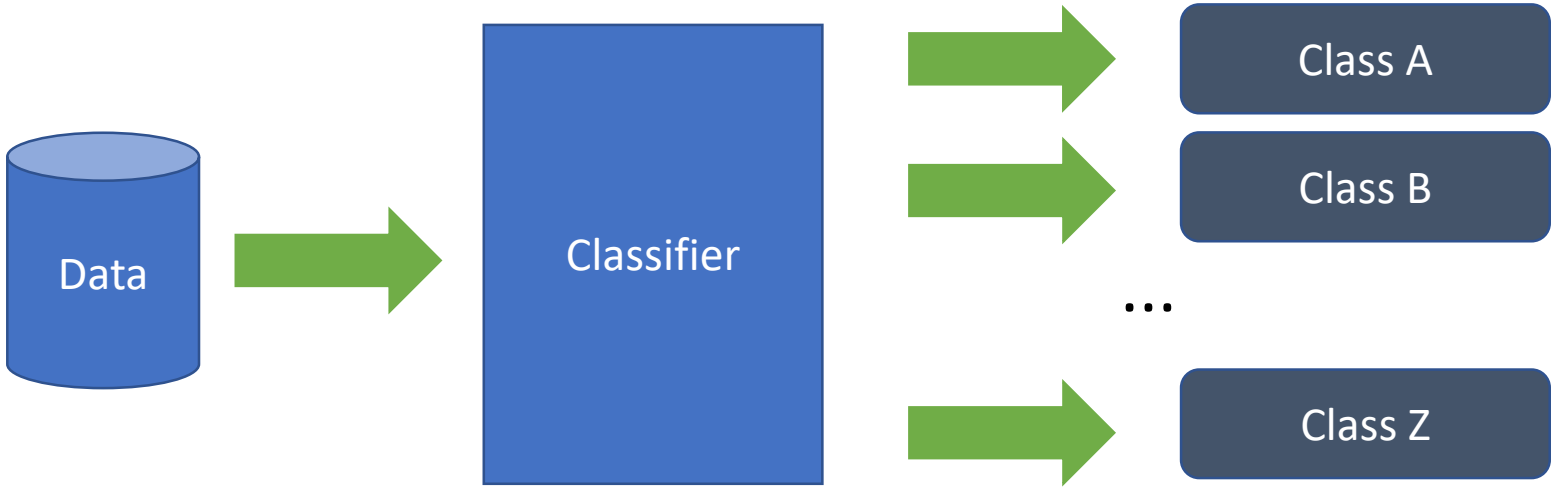
$$\mathrm{recall} = \frac{TP}{TP + FN}$$

ROC Area Under the Curve (ROC AUC)

# Multiclass Classification

- So far we have considered binary classifiers

- Multiclass classification



# Multiclass Classification

- Native multiclass classifiers
  - SGD
  - Random Forest
  - Bayes classifiers

- It is possible to use binary classifiers as multiclass classifiers
  - One-versus-the rest (OvR) (aka one-versu-all)
  - One-versus-one (OvO)

# One-Versus-The Rest (OvR)

- Build binary classifiers one for each class versus the other classes
- Usage
  - Apply data to all the classifiers
  - Select the class from classifier with the highest score

- Requires N classifiers for N classes

# One-Versus-One (OvO)

- Build binary classifiers one for each pair of classes
- Usage
  - Apply data to all the classifiers
  - Select the one with the highest score

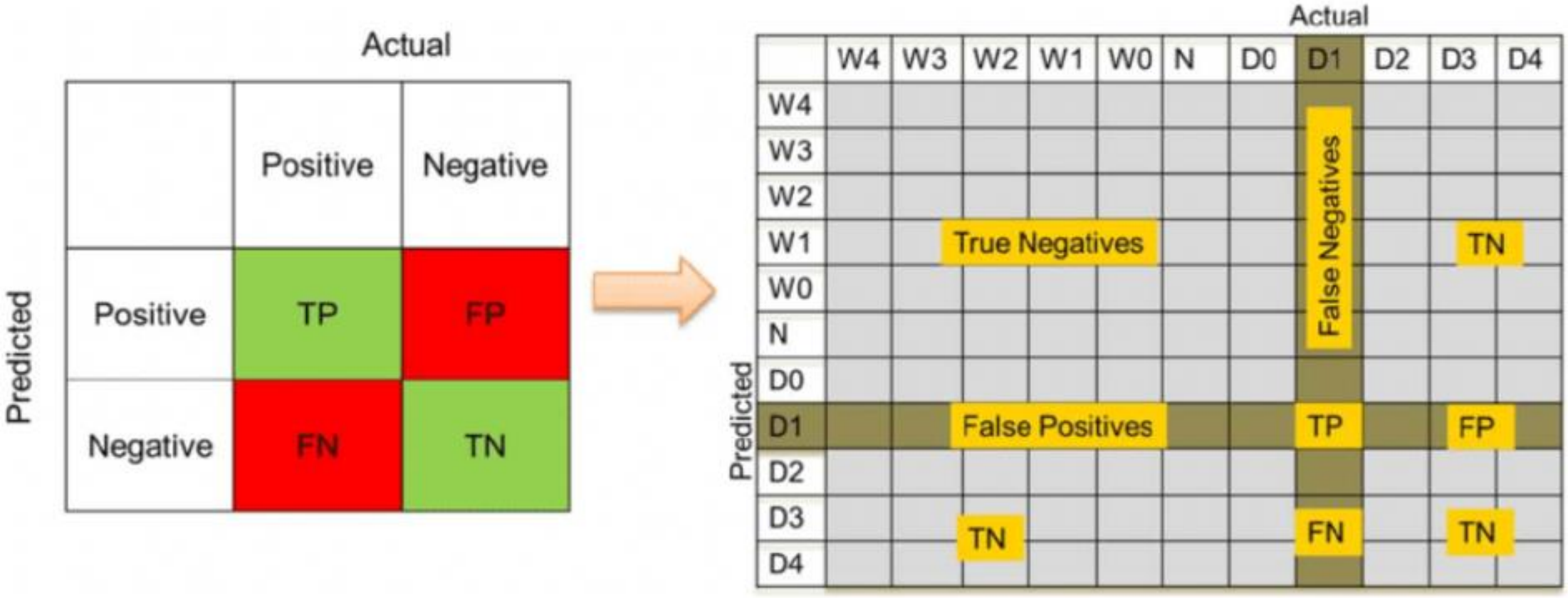
- N classes require N x (N-1)/2

# OvR vs OvO

- Size of the training set
  - OvO requires smaller training sets than OvR
  - E.g. SVM classifiers do not scale well with the size of the training set

- For most binary classifiers OvR is the preferred method

# Multiclass Confusion Matrix




https://www.researchgate.net/figure/Confusion-matrix-for-the-evaluation-of-binary-and-multiclass-drought-prediction-models_fig4_357906238

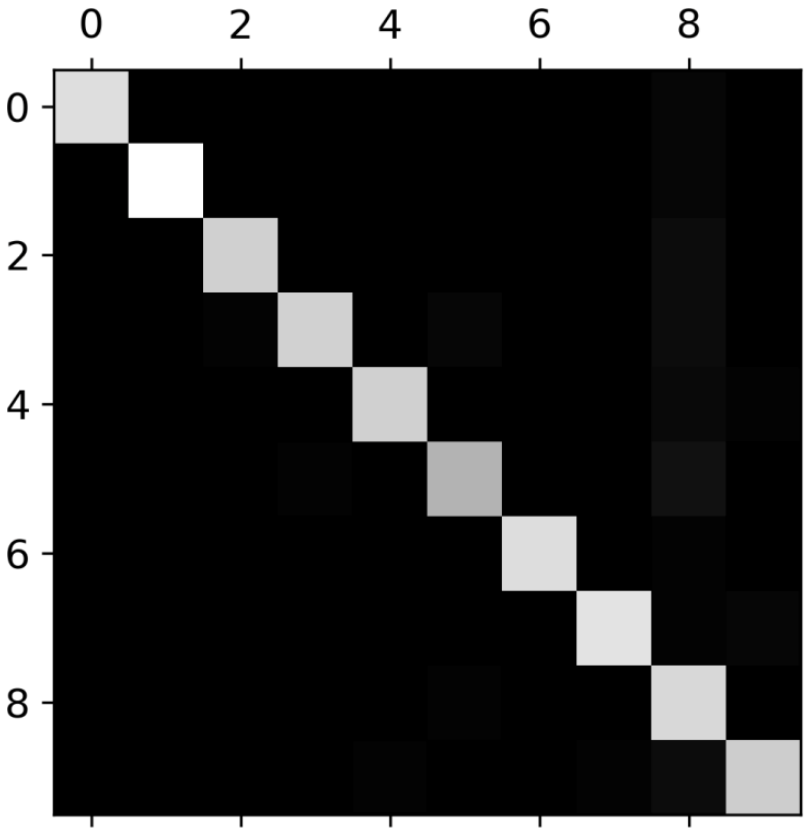
# Error Analysis

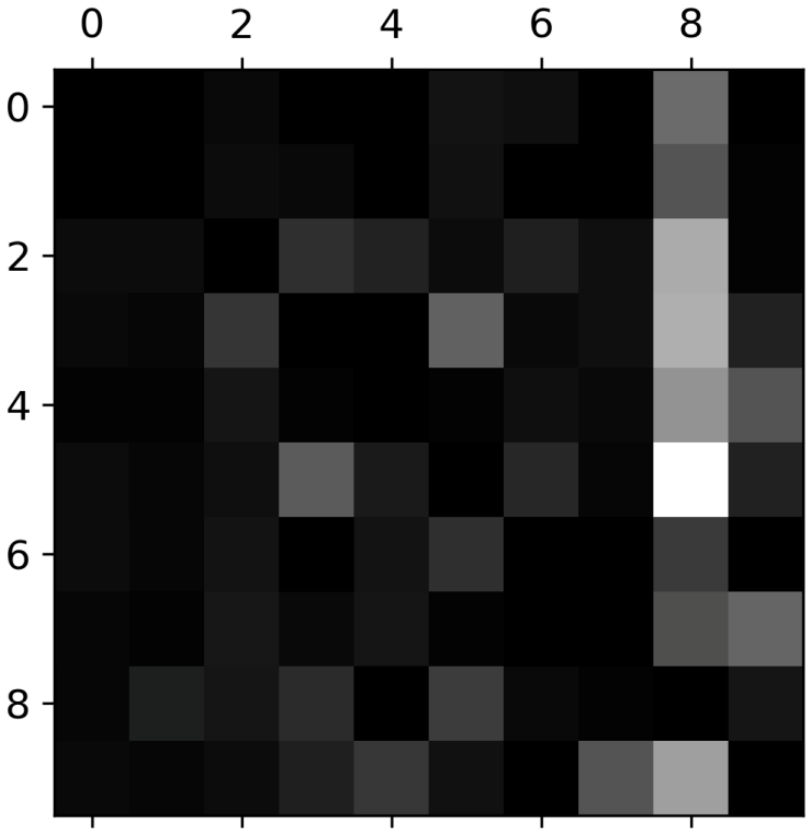
- The process of improving a ML model by reducing its errors
- Confusion matrix of SGD classifier on the MNIST digit database:

```
array([[5578,    0,   22,    7,    8,   45,   35,    5,  222,    1],
       [   0, 6410,   35,   26,    4,   44,    4,    8,  198,   13],
       [  28,   27, 5232,  100,   74,   27,   68,   37,  354,   11],
       [  23,   18,  115, 5254,    2,  209,   26,   38,  373,   73],
       [  11,   14,   45,   12, 5219,   11,   33,   26,  299,  172],
       [  26,   16,   31,  173,   54, 4484,   76,   14,  482,   65],
       [  31,   17,   45,    2,   42,   98, 5556,    3,  123,    1],
       [  20,   10,   53,   27,   50,   13,    3, 5696,  173,  220],
       [  17,   64,   47,   91,    3,  125,   24,   11, 5421,   48],
       [  24,   18,   29,   67,  116,   39,    1,  174,  329, 5152]])
```

# Error Analysis

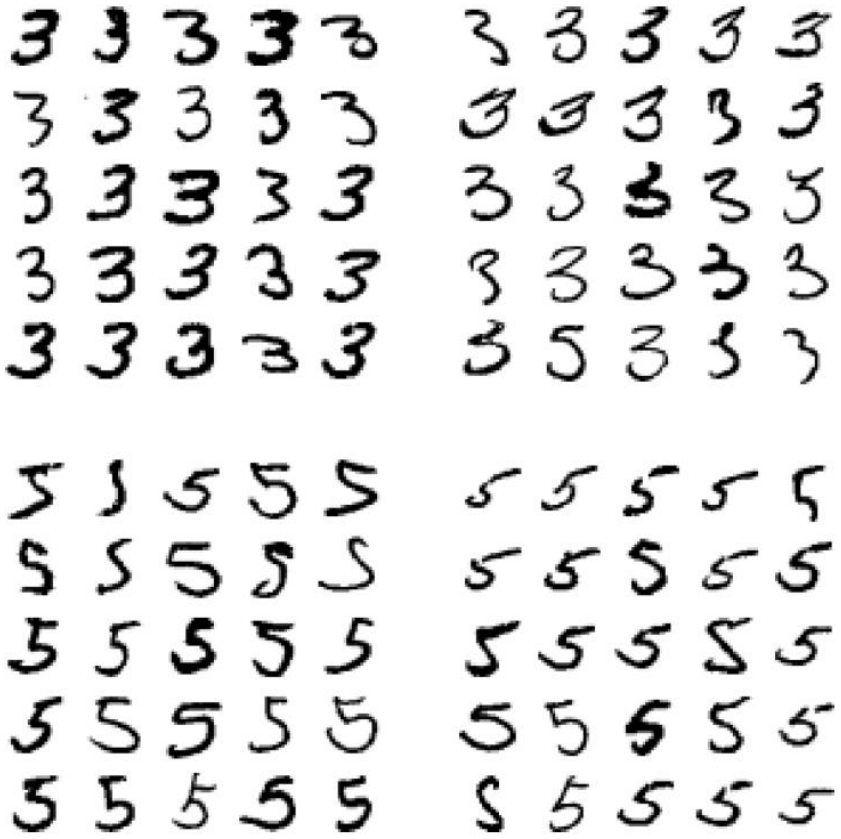
- Confusion matrix – graphical representation



# Error Analysis

- 3's and 5's are being confused for each other – let's take a look:

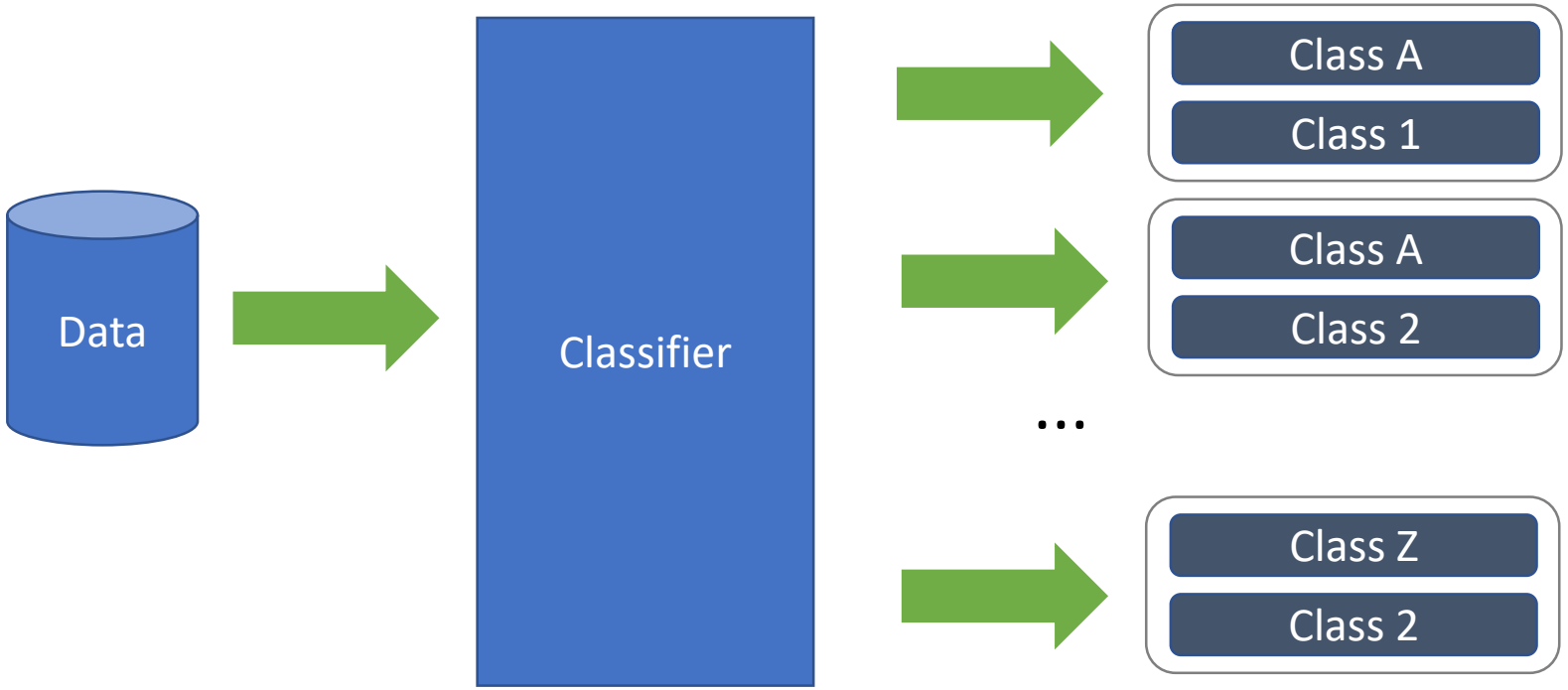


SGDClassifier is a linear model, and 3's and 5's often differ only by a few pixels.

Possible solution: preprocess the images so that they are perfectly centered.

# Multilabel Classification

- A data instance assigned to multiple classes



# Multioutput Classification

- A generalization of multilabel classification
- Each label can be multiclass
- Example
  - A classifier identifying if a pixel is noise or a part of a digit image
    - The output consists of pixels each being a label and a value (the intensity)